

| <b>PROJECT</b>                                                     |                |                                                                                    |                                                    | <b>JOB NO.</b> |                     |         |  |
|--------------------------------------------------------------------|----------------|------------------------------------------------------------------------------------|----------------------------------------------------|----------------|---------------------|---------|--|
| <b>CONTRACTOR :</b>                                                |                |                                                                                    |                                                    |                |                     |         |  |
| <b>SUB-CONTRACTOR : Winner Engineering Pte Ltd</b>                 |                |                                                                                    |                                                    |                |                     |         |  |
| <b>TRADE</b>                                                       |                | : ACMV SERVICES                                                                    |                                                    |                | <b>DRAWING NO :</b> |         |  |
| <b>PRE- TESTING COMMISSIONING CHECKLIST - VARIABLE SPEED DRIVE</b> |                |                                                                                    |                                                    |                |                     |         |  |
| <b>Unit Designation :</b>                                          |                |                                                                                    |                                                    |                |                     |         |  |
| <b>Unit Make :</b>                                                 |                |                                                                                    |                                                    |                |                     |         |  |
| <b>Model no :</b>                                                  |                |                                                                                    |                                                    |                |                     |         |  |
| <b>Location :</b>                                                  |                |                                                                                    |                                                    |                |                     |         |  |
| <b>Serial no :</b>                                                 |                |                                                                                    |                                                    |                |                     |         |  |
| S/N                                                                | Section        | Inspection / Test item                                                             | CHECKED CONDITION<br>(Please tick inside checkbox) |                |                     | Remarks |  |
|                                                                    |                |                                                                                    | Yes                                                | No             | NA                  |         |  |
| 1                                                                  | Documents      | Verify VSD tag, model, motor duty, ratings and approved wiring diagram.            |                                                    |                |                     |         |  |
| 2                                                                  | Installation   | Inspect enclosure, mounting, door seals and fan guards.                            |                                                    |                |                     |         |  |
| 3                                                                  | Cooling        | Check ventilation paths, filters and cooling-fan installation.                     |                                                    |                |                     |         |  |
| 4                                                                  | Electrical     | Verify panel, door, motor and cable-screen earthing.                               |                                                    |                |                     |         |  |
| 5                                                                  | Electrical     | Review cable and motor insulation-resistance test results.                         |                                                    |                |                     |         |  |
| 6                                                                  | Electrical     | Check isolated incoming and motor cables, terminations and phase identification.   |                                                    |                |                     |         |  |
| 7                                                                  | Electrical     | Verify isolator, protective devices and cable ratings.                             |                                                    |                |                     |         |  |
| 8                                                                  | Controls       | Check control cables, shielding, digital/analogue inputs and communication wiring. |                                                    |                |                     |         |  |
| 9                                                                  | Bypass         | Check bypass circuit and mechanical/electrical interlocks if fitted.               |                                                    |                |                     |         |  |
| 10                                                                 | Readiness      | Confirm driven equipment, guards, valves/dampers and operating path are ready.     |                                                    |                |                     |         |  |
| 11                                                                 | Startup        | Record incoming voltage and phase balance before run command.                      |                                                    |                |                     |         |  |
| 12                                                                 | Parameters     | Enter and verify motor nameplate data and application parameters.                  |                                                    |                |                     |         |  |
| 13                                                                 | Parameters     | Check min/max frequency, ramps, current limit, protection and restart settings.    |                                                    |                |                     |         |  |
| 14                                                                 | Controls       | Test local/remote start-stop, direction and run indication.                        |                                                    |                |                     |         |  |
| 15                                                                 | Cooling        | Verify drive cooling fans and ventilation during operation.                        |                                                    |                |                     |         |  |
| 16                                                                 | Speed test     | Test operation at 25 Hz where permitted by the load                                |                                                    |                |                     |         |  |
| 17                                                                 | Controls       | Test analogue reference, feedback scaling and PID response where used.             |                                                    |                |                     |         |  |
| 18                                                                 | BMS            | Verify BMS commands, speed feedback, run/trip status and alarms.                   |                                                    |                |                     |         |  |
| 19                                                                 | Protection     | Test relevant interlocks, emergency stop/STO and fault reset where fitted.         |                                                    |                |                     |         |  |
| 20                                                                 | Bypass         | Test VSD/bypass changeover where fitted using approved procedure.                  |                                                    |                |                     |         |  |
| 21                                                                 | Power recovery | Test communication-loss and controlled power-restoration response.                 |                                                    |                |                     |         |  |
| 22                                                                 | Performance    | Run at design duty; check motor current, noise, vibration and drive temperature.   |                                                    |                |                     |         |  |
| 23                                                                 | Handover       | Restore normal mode, save final parameters and close defects.                      |                                                    |                |                     |         |  |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |                     |  |
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| <b>Comments to follow up Action :</b><br><br><div style="border-bottom: 1px solid black; height: 15px; margin-bottom: 5px;"></div> <div style="border-bottom: 1px solid black; height: 15px; margin-bottom: 5px;"></div> <div style="border-bottom: 1px solid black; height: 15px; margin-bottom: 5px;"></div> <div style="border-bottom: 1px solid black; height: 15px; margin-bottom: 5px;"></div> <div style="border-bottom: 1px solid black; height: 15px; margin-bottom: 5px;"></div> |  |                     |  |
| Yes = satisfactory; No = rectify and re-inspect; NA = not applicable (state reason). Blank = not checked.                                                                                                                                                                                                                                                                                                                                                                                  |  |                     |  |
| <b>CHECKED BY:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  | <b>WITNESSED BY</b> |  |
| <b>NAME</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |                     |  |
| <b>DESIGNATION :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |                     |  |
| <b>COMPANY :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |                     |  |
| <b>SIGNATURE :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |                     |  |
| <b>DATE :</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |                     |  |